

User Query

1. API Gateway (Nginx)
 - Rate Limit Check (10 req/min per user)
 - Route to Java Backend

2. Java Backend (Spring Boot)
 - JWT Authentication Check
 - Check Redis Cache: Query → Response?

YES

Return Cached
Response

NO

3. Call Python AI Service (REST/gRPC)

4. Python AI Service
 - LangGraph Agent
 - a. Retrieve (Hybrid)
 - b. Grade Docs
 - c. Generate Answer
 - d. (Retry if needed)

5. Response Processing
 - Add Citations
 - Confidence Score
 - Save to Cache (Redis)
 - Log to PostgreSQL

6. Return to User
 - Answer + Sources
 - Metadata + Confidence